

CELIA RUBIO-MADRIGAL

 celrm.github.io

 celia.rubio-madrigal@cispa.de

 LinkedIn

 Google Scholar

I study ML generalization for graph and structured data, focused on how input structures shape data representation and computation. I have been a "la Caixa" Fellow, a 9th and 13th HLF attendee, and IT at EGMO since 2023.

EXPERIENCE



- **Doctoral researcher** – CISPA Helmholtz Center for Information Security **2023-now**
– Graphs, generalization, efficiency, structure awareness – Advisor: Dr. Rebekka Burkholz  
- **RA: Departmental collaboration** – Ministerio de Educación & Universidad Complutense de Madrid **2021-22**
– "Identifying Boolean functions that need big circuits with ML"  – Advisor: Dr. Ismael Rodríguez Laguna 
- **REU: Introduction to research ICMAT-Severo Ochoa** – Instituto de Ciencias Matemáticas (ICMAT) **2021**
– "Differential geometry methods in ML applications"  – Advisor: Dr. David Martín de Diego 
- **Intern: Consultant Developer** – Guidewire Software – NLP search engine to find internal raised issues. **2021**

EDUCATION



- **PhD in Computer Science** – CISPA & Universität des Saarlandes (Saarbrücken, Germany) **2023-now**
- **MSc in Advanced CS with AI** – University of Strathclyde (Glasgow, UK) – 8.57/10 (Distinction) **2022-23**
– Recipient of the "la Caixa" Fellowship  – Department's Best Overall Performance Prize (Top 1)
- **BSc in Mathematics** – Universidad Complutense de Madrid (Spain) – 9.08/10 **2017-22**
- **BSc in Computer Science** – Universidad Complutense de Madrid (Spain) – 9.16/10 **2017-22**

PUBLICATIONS



Legend: **bold**: First author in A* conference. * Equal contribution. † Co-supervision.

- R-M., Burkholz. *Linear GCNs Need Better Bias, Not More Expressive Power*. GFM Workshop @ ICML'26. 
- Joshi, Gadhikar[†], R-M.[†], Dutta, Singh, Burkholz. *Prune to Protect: Faster Training and Enhanced Privacy by Dynamic Data Pruning*. MemFM Workshop @ ICML'26. 
- R-M., Burkholz. *Fixed Aggregation Features Can Rival GNNs*. **ICML'26**. 
- Reddy, Verma, R-M., Muandet, Burkholz. *Boosting for Predictive Sufficiency*. ICLR'26. 
- Reddy, R-M., Burkholz, Muandet. *When Shift Happens - Confounding Is to Blame*. ICLR'26. 
- Jacobs, Gadhikar, R-M., Burkholz. *Hyperbolic Aware Minimization: Implicit Bias for Sparsity*. ICLR'26. 
- R-M.*, Jamadandi*, Burkholz. *GNNs Getting Comfy: Community and Feature Similarity Guided Rewiring*. **ICLR'25**. 
- Jamadandi*, R-M.*, Burkholz. *Spectral Graph Pruning against Over-squashing and Over-smoothing*. **NeurIPS'24**. 

AWARDS



- 13th Heidelberg Laureate Forum: 200 international young researchers network with math and CS laureates **2026**
- Dept. Best Overall Performance Prize: University of Strathclyde's top student in MSc Advanced CS **2023**
-  Nova III Student List: Spain's top 10 talents under 25 in Computer Science **2023**

- 9th Heidelberg Laureate Forum. Featured in: 📺 *HLFF Spotlight*, 📺 *Tribuna Complutense* 2022
- 🎧 "la Caixa" Fellowship: *Full funding for MSc studies*. Featured in: 📺 *El País*, 📺 *El Plural*, 📺 *TeleMadrid* 2022
- **Undergraduate**
 - 🎧 Aula de Verano "Blas Cabrera" for Spain's top undergrad records: *Intl. Univ. Menéndez Pelayo (UIMP)* 2022
 - 🎧 SWERC contestant (competitive programming): *ICPC Southwestern Europe Regional Contest in Italy* 2022
 - Becas de Excelencia: *Excellence Grants for undergrads with >9/10 in Madrid (2100€)* 2018-22
- **Pre-undergraduate**
 - 🎧 Premio Extraordinario de Bachillerato: *Madrid's top 25 in general-content exam (1000€)* 2017
 - 🎧 EGMO contestant representing Spain: *European Girls' Mathematical Olympiad in Switzerland* 2017

TALKS



- **Invited talks** – *Seminars and reading groups* 2025-26
 - 🎧 GLOW, 🎧 TUWien ML RU seminar, 📺 Cohere For AI, CISPA ML seminar, 🎧 AI Security Afternoon.
- **Contributed talks** – *Conferences and workshops* 2024-26
 - GSP 🎧 2026, HAICON 🎧 2024. Lightning talks: GTML 🎧 2025, MLG 🎧 2025 (2), NetSci 🎧 2025.
- **Non-archival posters** – *Non-archival conferences, workshops, and sessions* 2024-26
 - GSP 🎧 2026, CPAL 🎧 2026 (3), WiML 🎧 2024 🎧 2025, ELLIS Pre-NeurIPS/ICML 🎧 2024 🎧 2025 🎧 2026.

TEACHING



- **Math Olympiad Trainer** – 🎧 *International EGMO Preparatory Seminar in Bonn, Germany* 2026
- **Master's and Bachelor's thesis advising** – *CISPA & Universität des Saarlandes (3)* 2024-now
- **Teaching Assistant: Laboratory demonstrator (7h/week)** – *University of Strathclyde* 2023
 - "CS310 Foundations of AI" for Year 3 BSc Computer Science – Lecturer: Dr. Andrew Abel 📺

SERVICE



- **Academic service** 2024-now
 - Reviewer: MLG 2025-26, ICLR 2026 – Top 200 (1%), CPAL 2026, ICML 2026 – Gold (25%), NeurIPS 2026.
 - Volunteer at 19th and 20th Women in Machine Learning Workshop (WiML) at NeurIPS 2024 and 2025.
- **Math Olympiad service**
 - 🎧 IT at European Girls' Mathematical Olympiad (EGMO) with Joseph Samuel Myers 📺 2023-now
 - Deputy Leader of Spain at EGMO 2022 and PAGMO 2021. 2021-22
 - Jury at the local phases of 57th to 61st Olimpiada Matemática Española (OME) Madrid. 2020-24
 - Guide at the Iberoamerican Mathematical Olympiad 2018.
- **Student representation** – *Facultad de Informática (Fdi), Universidad Complutense de Madrid* 2020-22
 - Secretary of the Fdi Student Council 2021-22. Member of the Fdi Faculty Board 2020-22.
- **L^AT_EX package development: ddphonism** 2019
 - 🎧 Twelve-tone music notation diagrams with TikZ. Included in MiKTeX and TeXLive (try it in Overleaf).
- **Mentoring and outreach**
 - Mentor for new "la Caixa" Fellows 2023-25
 - Outreach articles for DivulgaMAT (RSME journal): *Serialismo y matemáticas* 📺 📺 📺 📺 2019
 - Outreach videos for the Spanish math education community: 📺 📺 📺 2018-21